

INVESTIGATION REPORT

Issue 993: Does CV32E40X fetch its first instruction from boot+4?

openhwgroup/cv32e40x

Document	RTL-TRIAGE-CV32E40X-993
Disposition	Not reproduced
Record date	2026-09-11
Classification	Internal working document. Not a verification sign-off.

1 Status

Issue	openhwgroup/cv32e40x#993
Issue type	Bug report from an integrator or user
Affects	cv32e40x
Configuration	X_EXT=0 (default), RTL master
Disposition	Not reproduced
Severity	Not assessed
Agreement	No independent human record was available for comparison.
Evidence	5 cited, 3 present, 2 not retained

Disposition meaning. The reported behaviour did not occur under the conditions investigated. This is not a statement that no defect exists.

Severity is not recorded for this run. The harness does not infer a severity from the disposition, so the field is reported as not assessed rather than filled in.

2 Description

The reported behaviour (first accepted instruction fetch after reset at boot_addr_i + 4) does not occur at the cv32e40x_core boundary under a quiet environment. With OBI handshake legality, reset released once, boot_addr_i stable, and no debug request / interrupt / bus error until the first fetch is accepted, an unbounded proof (abc pdr) plus a bounded check over 30 cycles (the first fetch is accepted in cycle 3) show the first accepted instr_addr_o equals {boot_addr_i[31:2],2'b00}, and a cover for 'first accepted fetch == boot+4' is unreachable in that window. Without the quiet-environment assumption the solver does produce a counterexample, but it is an environment artifact: it asserts debug_req_i while the core is in RESET, so the core enters DEBUG_TAKEN and its first fetch goes to dm_halt_addr_i (a free input the solver set to boot+4). The most likely explanation of the reporter's observation is therefore something in their integration (a debug halt or interrupt at boot, or a monitor sampling instr_addr_o one cycle after gnt), not the core. Recommended: ask the reporter for debug_req_i/irq_i/dm_halt_addr_i values at reset and the exact cycle of gnt relative to instr_addr_o.

3 Conditions

Not recorded as separate fields for this run. The conditions and environment assumptions that bound this result are stated inside the Description, and the configuration is given in Status.

4 Implications

Not recorded as a separate field for this run. What the disposition means for a user of the core follows from the Disposition meaning in Status and the bounds in Limitations and scope.

5 Action

Checker used

```
checker_993_quiet.sv
```

6 Evidence

Ref	Type	Path, relative to the run directory	File	Result recorded
E1	proof	formal/prove_quiet/sby.log	present	abc pdr unbounded PASS (quiet environment): first accepted fetch after reset == boot address, for every input sequence
E2	proof	formal/bmc_quiet/sby.log	present	BMC depth 30, quiet environment: assertion 'first accepted fetch == boot address' PASS (26 s)
E3	proof	formal/cover_quiet/sby.log	present	cover 'first accepted fetch' REACHED at step 3; cover 'first accepted fetch == boot+4' UNREACHED to depth 30
E4	counterexample	formal/bmc_fixedboot/job/engine_0/trace.vcd	not kept	Unconstrained-debug counterexample: debug_req_i during RESET -> DEBUG_TAKEN -> first fetch to dm_halt_addr_i. Explains how a 'wrong first fetch' can be observed; rendered in cex_env_artifact.png
E5	counterexample	formal/cover_quiet/job/engine_0/trace0.vcd	not kept	witness of the normal boot: fetch_branch to boot at cycle 2, accepted request at boot address cycle 3; cover_quiet_first_fetch.png

Paths are relative to runs/e40x-993. The File column is a check made when this document was generated. It reports whether the cited file exists, not whether its contents were re-validated. 'not kept' means a repository rule deliberately excludes that artefact, usually a waveform dump; rerun the check to regenerate it. 'MISSING' means the file is absent and was expected to be there.

What each class of artefact establishes

- proof. Proof or engine run. Scope depends on the mode: prove is unbounded under the stated model, bmc is bounded to its depth, cover establishes reachability only.
- counterexample. A concrete input sequence produced by the engine. Shows the stated condition is reachable; it is not itself a proof.

Artefact(s) E4, E5 are not retained under repository policy. The result recorded against them stands on the engine log cited alongside; the artefact itself must be regenerated by rerunning the check before it can be inspected.

7 Limitations and scope

- Absence of reproduction is not proof of absence of a defect. A different configuration, stimulus or tool version may still expose the reported behaviour.
- This is an agent investigation record. It requires engineering review before it is used as a disposition. No report generator can supply missing evidence.
- Results apply to the configuration and core revision named in Status. A different configuration needs a separate run.
- Measured tool time is the time the tools ran. It is not an engineering effort figure and not a turnaround commitment.
- Public demonstration runs on open cores are not client engagements and not production verification results.

8 Provenance and measurement

Run directory	runs/e40x-993
Title source	Title and configuration read back from reports/issues/cv32e40x_issue_993.pdf, the previously published report for this issue.
Inputs shown	title, body
Human record	open issue, no maintainer response yet (as of 2026-09-10)
Agent confidence	0.9. This is the investigating agent's own estimate. It is not a measurement and not a

reliability figure.

Iterations5

Run byClaude directly

Measured tool and step time

Step	Measured
fetch + read RTL (if_stage, prefetcher, controller)	10.0 min (600 s)
checker v1 + cover/BMC	40.0 s
diagnose artifact (async boot_q, re-reset) v2/v3	5.0 min (300 s)
fixed-boot BMC + trace dump	60.0 s
diagnose DEBUG_TAKEN artifact, quiet checker	5.0 min (300 s)
quiet BMC + cover	56.0 s
report	10.0 min (600 s)
unbounded pdr proof	21.0 s
Total	32.6 min (1956 s)

Times are wall-clock for the step named. Some steps may have run concurrently, so they do not sum to an isolated performance figure, and they are not an engineering effort measure.

Generated by reporting/scripts/issue_document.py from runs/e40x-993/verdict.json. The generator typesets recorded data and checks that cited files exist. It does not run, rerun or validate any check, and it cannot supply missing evidence.